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Hypolipidemic and Antioxidative Effects of *Glossogyne tenuifolia* in Hamsters Fed an Atherogenic Diet

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ABSTRACT *Glossogyne tenuifolia* (GT) Cassini is a special herbal tea in the Penghu Islands, Taiwan, and has a long history of being used as an antipyretic, detoxifying, and anti-inflammatory remedy in folk medicine among local residents. The aim of this study was to investigate the effect of hot water extracts from GT on oxidative stress and lipid metabolism in animals. Five- to 6-week-old male Syrian hamsters were divided into four groups ($n = 14$) for different treatments, that is: control group (C), high-fat/cholesterol (HF) group, HF diet containing 0.5% (GT0.5) and 1.5% (GT1.5) GT extracts for 4 weeks. Hamsters fed with 0.5% GT powder as well as 1.5% GT powder exhibited reduced serum total cholesterol (TC), conjugated diene of low-density lipoprotein (LDL), and increased serum antioxidant capacity, but 1.5% GT powder was more potent at lowering serum LDL cholesterol and thiobarbituric acid reactive substance concentrations than 0.5% GT. GT extracts significantly lowered liver triacylglycerol (TG) concentration by diminishing activities of fatty acid synthase (FAS) and glucose-6-phosphate dehydrogenase (G-6-PDH). In addition, fecal excretion of cholesterol and bile acids were increased in GT extract groups. In conclusion, GT extracts increase the antioxidative capacity, decrease serum TC, inhibit the activities of FAS and G-6-PDH, and further reduce liver TG accumulation in hamster fed on atherogenic diets.

KEY WORDS: • antioxidant capacity • bile acid • *Glossogyne tenuifolia* • hamsters • hypolipidemic

INTRODUCTION

It is well known that hypercholesterolemia appears critical to the atherogenic process and tends to lead to serious cardiovascular disease (CVD). Over the past several decades, CVD has become the leading causes of death in developing and developed countries. The oxidative hypothesis of atherosclerosis suggests that circulating low-density lipoprotein (LDL) is oxidized *in vivo*, which leads to its enhanced uptake by macrophages inside the arterial wall, and is believed to subsequently result in foam cell formation, one of the first stages of atherogenesis.[1,2] Therefore, suppressing LDL oxidation would be helpful in lowering the incidence of CVD. On the other hand, antioxidants play important roles in preventing lipid oxidation and slowing the progression of atherosclerotic lesions.[3]

The health-promoting properties of some herbal teas or plants reported to be rich in antioxidants, particularly phenolics and antioxidant vitamins, can exert protective action against oxidative stress and exert antihyperlipidemic effects.[4–6] *Glossogyne tenuifolia* (GT) Cassini, also called Hsiang Ju grass, is a perennial herb with woody stems at the base, which is distributed mainly in exposed coastal areas of Penghu Island, Taiwan. It is traditionally used to make a healthy herb tea for preventing sunstroke and has a long history of being used as an antipyretic, hepatoprotective, and anti-inflammatory remedy in folk medicine among local residents. However, there is still limited scientific evidence about the biological functions of GT as a nutraceutical beverage. Some bioactive components from GT have been isolated and shown to exert combinatorial anti-inflammatory and antiviral effects in the production of some mediators.[7] In addition, GT exhibits a potential reactive oxygen species scavenger effect and may prevent atherosclerosis through inhibiting LDL oxidation *in vitro*.[8] Recently, we showed that hot water extract of GT was rich in phenolic compounds, for example, chlorogenic acid (CGA), and exhibited hepatoprotective and antioxidant activities in an animal study.[9] The roles and applications of CGA, particularly in relation to glucose and lipid metabolism, have been highlighted.[10,11] However, it is still unclear whether GT exhibits a hypolipidemic effect and retards LDL oxidation in an animal model. Therefore, the present study was carried out to investigate whether the hot water extract of GT exerts antihyperlipidemic and antioxidative effect in hamsters fed on high-fat/cholesterol (HF) diets.

MATERIALS AND METHODS

Animals and diets

GT Cassini (Hsiang Ju grass) was obtained and its hot water extracts were prepared as previously described.[9]

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Six-week-old male Syrian hamsters were obtained from the National Laboratory Animal Center (Taipei, Taiwan) and fed a laboratory rodent diet for about 7 days until their body weight reached a range of 70–80 g. Fifty-six animals were divided into 4 different dietary groups of 14 animals each. The animals were given free access to all diets and water. The control diet was maintained as the initial rodent diet, while the atherogenic diet was the HF diet derived from an adjusted calorie diet (TD 88137; Harlan Laboratories, Indianapolis, IN, USA) containing 21% milk fat and 0.2% cholesterol. The experimental diets were HF diets containing 0.5% and 1.5% GT extracts (GT0.5 and GT1.5), respectively.

All hamsters were housed two each in stainless steel cages and kept in a room maintained at 22–24°C with controlled lighting in a12-h light–12-h dark cycle. Food intake and body weight were recorded daily and weekly, respectively. All animal experimental procedures were approved by the Animal Experimental Committee of the Fu Jen University.

Sample preparation

At the end of the experiment, animals were food deprived overnight (12–14 h) and killed under carbon dioxide anesthesia by withdrawing blood using a vacuum tube from the heart between 9:00 am and 11:00 am. Serum and liver handling and storage were performed as previously described.[12] The serum from each of the two hamsters with equal volume (about 1.2 mL) were pooled to make one sample for the following analysis.

Liver homogenate was prepared with 10 volumes of ice-cold 10 mM phosphate-buffered saline (pH 7.4) containing 1.15% KCl using a Potter–Elvehjem-type homogenizer. Portions of the homogenates were measured immediately for thiobarbituric acid reactive substances (TBARS). Fecal samples were collected for 2 days before sacrifice, lyophilized, and ground.

Lipid analyses and antioxidant capacities

Commercial enzymatic kits (Fortress Diagnostics, Belfast, United Kingdom) were used for analysis of serum total cholesterol (TC), low-density lipoprotein cholesterol (LDL-C), high-density lipoprotein cholesterol (HDL-C), and triacylglycerol (TG). Hepatic and fecal lipids were extracted using the procedure developed by Folch *et al.*[13] For determination of liver TC and TG, and fecal TC, an aliquot of the extract was dissolved in Triton X-100.[14] TC and TG were then determined using enzymatic reagent kits as used in serum.

LDL was isolated from serum containing EDTA using buffered heparin as described by Ahotupa *et al.*[15] and the lipids of the LDL pellet were extracted.[13] The organic layers were dried under nitrogen and redissolved in 1 mL of cyclohexane. The conjugated diene concentration was measured spectrophotometrically at 234 nm. The results were expressed as micromoles per liter of serum using the molar extinction coefficient 2.95×10^4 M^{-1} cm^{-1}.

Serum and hepatic TBARS were measured using tetraethoxypropane as the standard.[16] The total antioxidant capacity was measured,[17] the absorbance was measured at 734 nm and expressed as Trolox equivalent antioxidant capacity (TEAC) as described previously.[9] For determining fecal bile acids, a small aliquot of feces was extracted with anhydrous alcohol and determined by a commercial kit (Randox Laboratories, Antrim, United Kingdom) using taurocholic acid as the standard.

Activity of hepatic lipogenic enzymes

A 0.5 g portion of frozen liver was homogenized in 10 volumes of an ice-cold 0.25 M sucrose solution containing 1 mM EDTA in 10 mM Tris-HCl buffer (pH 7.4). The homogenate was centrifuged at 98,000 *g* for 10 min and the supernatant was recentrifuged at 105,000 *g* for 60 min to precipitate microsomes and the remaining supernatant was used for determination of enzyme activities.

Enzyme activities were measured spectrophotometrically at 340 nm for 4 min at 30°C with a UV-1240 spectrophotometer (Shimadzu, Kyoto, Japan) and expressed on the basis of nanomole NADPH oxidized to NADP (or NADP reduced to NADPH) per minute per milligram protein using the extinction coefficient of NADPH at 340 nm of 6.22×10^6 M^{-1} cm^{-1}. Fatty acid synthase (FAS), glucose-6-phosphate dehydrogenase (G-6-PDH), and malic enzyme were measured according to the classic methods as previously described.[18]

Table 1. Effects of Different Dosages of GT Extracts on Serum Lipid and Lipoprotein Concentrations in Hamsters

	Groups			
Parameters	*Con, mg/dL*	*HF, mg/dL*	*GT0.5, mg/dL*	*GT1.5, mg/dL*
TG	110.8 ± 44.4^{a}	122.6 ± 9.1^{a}	125.5 ± 16.8^{a}	121.2 ± 15.0^{a}
TC	136.0 ± 27.5^{b}	172.6 ± 17.1^{a}	141.4 ± 23.8^{b}	135.5 ± 20.6^{b}
LDL-C	19.2 ± 11.0^{b}	34.2 ± 10.5^{a}	28.4 ± 8.8^{ab}	19.9 ± 8.6^{b}
HDL-C	94.4 ± 11.2^{b}	135.3 ± 19.5^{a}	127.5 ± 14.2^{a}	128.0 ± 17.6^{a}
LDL-C/HDL-C	0.21 ± 0.14^{ab}	0.25 ± 0.05^{a}	0.23 ± 0.06^{a}	0.16 ± 0.05^{b}

Values are mean ± standard deviation ($n = 6$–7).

[abc]Values in the same row with different letters are significantly different at $P < .05$.

GT, *Glossogyne tenuifolia*; HDL-C, high-density lipoprotein cholesterol; HF, high-fat/cholesterol; LDL-C, low-density lipoprotein cholesterol; TC, total cholesterol; TG, triacylglycerol.

Statistical analyses

The results are expressed as mean ± standard deviation in each animal group. Data were analyzed by one-way ANOVA followed by testing differences between pairs by Tukey's range test. $P<.05$ was regarded as statistically significant.

RESULTS

Influence on serum and liver lipids

The concentration of TC in hamsters fed on HF diet was markedly higher than that of hamsters fed on the control diet. The addition of either 0.5% or 1.5% GT extracts to the HF diet significantly decreased the serum TC concentration of hamsters (average ca. 18% lower), and the lowering effect was similar between GT0.5 and GT1.5 groups (Table 1). HF diet feeding also increased LDL-C and HDL-C, while the addition of 1.5% GT to HF diets decreased the concentration of LDL-C, but resulted in no change in HDL-C. The ratios of LDL-C/HDL-C in hamsters fed on 1.5% GT were lower than that of HF diets. On the other hand, the concentration of serum TG on hamsters fed on HF diets was similar to that of hamsters fed on the control diet, irrespective of GT addition.

Hepatic TG concentrations in hamsters fed on GT-containing diets were significantly and dose-dependently lower than those on the HF diet (Fig. 1). Hepatic TC of hamsters fed on HF diets was markedly higher than that of hamsters fed on the control diet. However, the addition of either 0.5% or 1.5% GT extract to the HF diet did not significantly decrease hepatic TC concentration in hamsters.

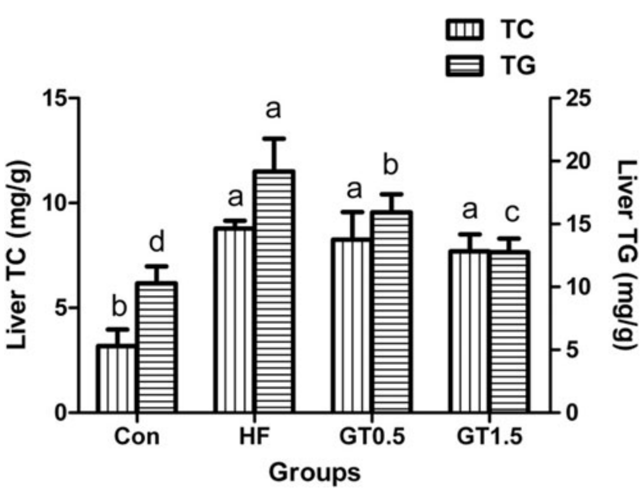

FIG. 1. Effects of different dosages of GT extracts on liver lipid concentrations in hamsters. Data are mean ± SD ($n=7$). Con, control group; HF, high-fat/cholesterol diet; GT0.5 and GT1.5 were HF diet containing 0.5% and 1.5% GT powders, respectively; TC, total cholesterol; TG, triacylglycerol. [abcd]Values with various *letters* are significantly different at $P<.05$. GT, *Glossogyne tenuifolia*; SD, standard deviation.

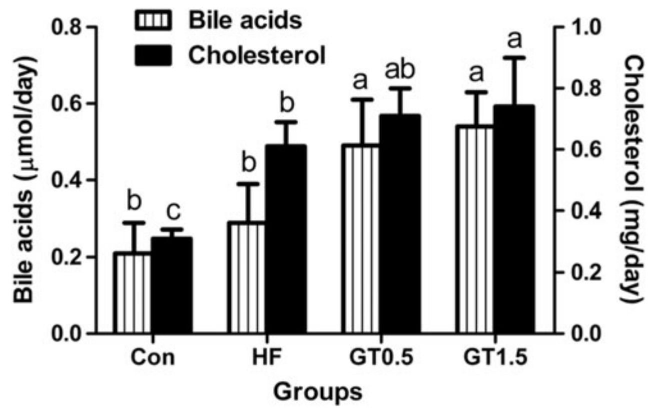

FIG. 2. Fecal bile acids and cholesterol contents of hamsters fed different dosages of GT. Data are mean ± SD ($n=7$). [abc]Values in the same *row* with different *letters* are significantly different at $P<.05$.

Influence of GT on fecal excretion

Hamsters fed on HF diets had increased fecal excretion of bile acids and cholesterol (Fig. 2); the addition of GT extracts to HF diets significantly increased fecal excretion of bile acids in comparison to HF diets irrespective of dose. In addition, the excreted contents of fecal cholesterol in GT1.5 group were higher than that of HF group, and were similar to that of the GT0.5 group.

Influence on liver enzyme activities

The lipogenic enzyme activities of hepatic FAS and G-6-PDH were lower in hamsters fed on GT1.5 diets than in hamsters fed on HF diets (Fig. 3). Moreover, the activity of hepatic malic enzyme tended to be lower in hamsters fed on GT1.5 diets than that of those fed on a corresponding HF diet.

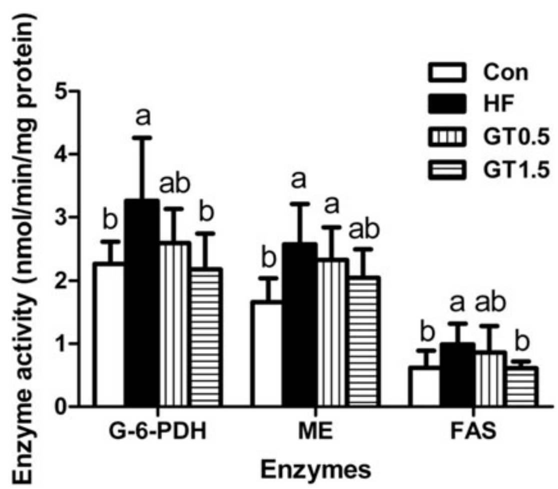

FIG. 3. Effects of different dosages of GT extracts on the activities of FAS, ME, and G-6-PDH in the liver. Data are mean ± SD ($n=7$). [abc]Values with various *letters* are significantly different at $P<.05$. FAS, fatty acid synthase; G-6-PDH, glucose-6-phosphate dehydrogenase; ME, malic enzyme.

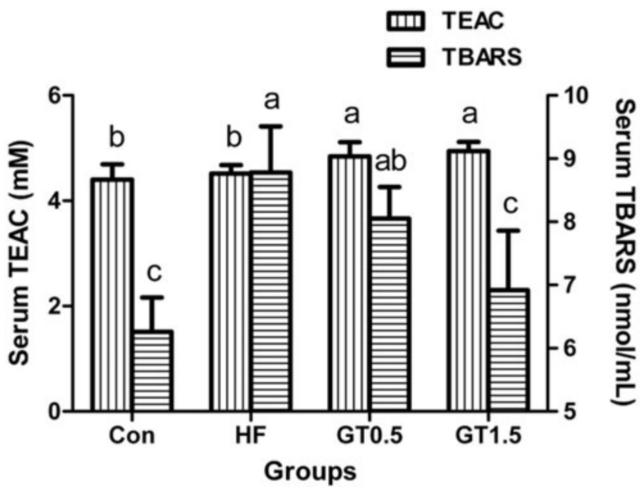

FIG. 4. Effects of different dosages of GT extracts on serum TEAC and TBARS in hamsters. Data are mean ± SD ($n=7$). TEAC, trolox equivalent antioxidant capacity; TBARS, thiobarbituric acid reactive substances. ^{abc}Values with various *letters* are significantly different at $P<.05$.

Antioxidant status of serum and LDL

The serum TBARS of hamsters fed on HF diet was significantly higher than that of hamsters fed on the control diet, and the addition of 1.5% GT extracts to HF diet decreased serum TBARS (Fig. 4). Furthermore, the serum TEAC of hamsters fed on HF diet were comparable with that of hamsters fed on control diet, and the addition of GT extracts to HF diet significantly increased serum TEAC.

Although TBARS of the serum LDL fraction in hamsters fed on HF diets with GT extracts were similar to those of the control group, hamsters fed on HF diet with GT extracts exhibited more effective retardation of the conjugated diene formation in LDL than in hamsters fed on HF diet (Fig. 5).

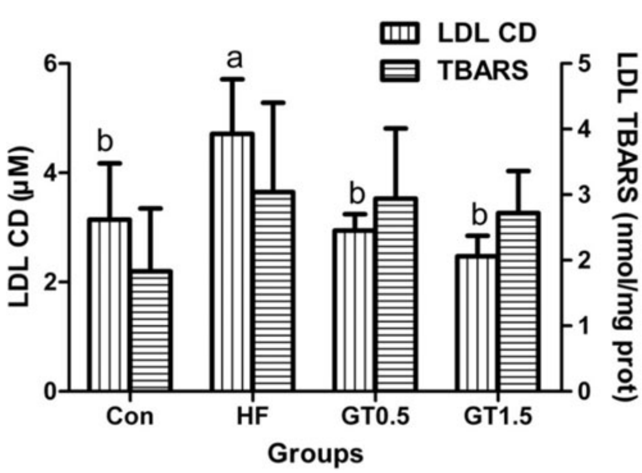

FIG. 5. Effects of different dosages of GT extracts on CD formation and TBARS of serum LDL in hamsters. Data are mean ± SD ($n=6$–7). ^{ab}Values with various *letters* are significantly different at $P<.05$. CD, conjugated dienes; LDL, low-density lipoprotein.

DISCUSSION

The Syrian hamster is a widely used animal model for lipid research, as its cholesterol and lipoprotein metabolism and transport are similar to those of humans. The aim of the present study was to evaluate whether GT exerts hypolipidemic and antioxidant effects on hamsters fed on a HF diet. In hamsters weighing an average of 76–78 g at initial body weight, the body weight gain was higher in hamsters fed on HF diet and HF diet with GT extracts than for the control. High-fat diet feeding increased the body weight as would be predicted, and was consistent with the previous results in hamsters.[18]

The hypocholesterolemic effects of GT might be attributed to the fact that GT extracts contain the phenolic compounds, CGA and luteolin-7-glucoside, as previously reported,[9] where both exert antilipidemic capacity in reducing cholesterol and TG concentrations in serum and liver,[19,20] and the former also inhibits liver HMG CoA reductase.[21] It has been shown that CGA improves lipid metabolism in mice fed on HF diet,[22] the phenomenon is similar to our results that hamsters fed on GT-containing diets significantly lowered hepatic TG compared to those on the HF diet.

Feeding GT to hamsters can stimulate fecal steroid excretion and might have mediated, at least in part, the hypocholesterolemic effect in the present experiment. Our study seemed to consistently show that the hypolipidemic effect correlated with the extent of fecal steroid excretion and lipogenesis in liver, suggesting that feeding hamsters a diet containing more GT would be more effective at preventing hyperlipidemia.

The major function of G-6-PDH and malic enzyme is to provide NADPH for reductive biosynthesis of fatty acids or cholesterol. The lower activities of FAS and G-6-PDH were consistent with the TG accumulation and NADPH production in our experiment, and the lower G-6-PDH activity was significantly correlated with liver total lipids.[23] In addition, our results also showed that hamsters fed on GT1.5 diet might experience reduced lipogenesis in comparison with hamsters fed on a HF diet.

Phenolics and flavonoids present in herbs and vegetables have received considerable attention due to their potential antioxidant activities. Although CGA *per se* exhibits antioxidant properties *in vitro*,[24,25] our results demonstrate that dietary supplementation with GT extracts significantly enhances the resistance to LDL oxidation and the antioxidant capacity of serum in hamsters. The GT feeding exhibits antihyperlipidemic action; enhancing antioxidant activities could potentially be highly effective for preventing CVDs.

In conclusion, the results of this study demonstrated that feeding HF diet containing GT extracts to hamsters might be effective for lowering serum TC and LDL-C and attenuating the cardiovascular risk, which can be attributed to the increases in fecal sterol and bile acid levels. The GT extracts also lowered hepatic TG contents, which can be attributed to the inhibition of lipogenesis in the liver. Moreover, GT extracts added to the diet increased the antioxidant activity by lowering TBARS in serum and conjugated diene formation in LDL fraction. The complementary actions of GT extracts may be beneficial for cardiovascular protection.

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AUTHOR DISCLOSURE STATEMENT

No competing financial interests exist.

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